

Python for Artificial Intelligence

Day 1: Orientation & Python Basics

Month 1: Python Mastery | LLM & Agentic AI Track

Today's Agenda

- Hour 1: Welcome, your 3-month roadmap, and Pakistan's AI/ML job landscape
- Hour 2: Environment setup, Python, VS Code, your first script
- 15-Minute Break
- Hour 3: Python fundamentals, variables, data types, operators



Hour 1: Welcome & Roadmap

Welcome to the Program

- 3 months, full-time: 5 days a week, 3-hour sessions, 195 total hours
- One specialization, mastered, not a survey of everything in AI
- You will leave with two real, working, published projects, not a list of topics you once saw
- Every week ends with a named skill you can claim; every month ends with a demo

Your 3-Month Roadmap

- Each month has one job: master Python, then build one real AI application, then ship and sell it
- Every month closes with a project sprint week and a Demo Day, no exceptions



one specialization, mastered, not many topics surveyed

How You'll Be Assessed

- Weekly Skill Checkpoint: every Friday, a short, specific test of that week's named skill
- Monthly Portfolio Milestone: the last week of every month has no new content, it's a project sprint
- Month 1 milestone: "I can build and ship a real Python program independently"
- Month 2 milestone: "I can build and deploy a working AI application using LLMs, RAG, and agents"
- Month 3 milestone: "I can ship a production-ready AI application and present it professionally"

Pakistan's AI/ML Job Landscape

Why this specific specialization, and why now:

- Pakistan's AI/ML market is projected to grow roughly 28% a year, toward around USD 3.2 billion by 2030
- In a 2025 industry survey, OpenAI-related and LLM API skills were the single most requested AI/ML tool skill in local job postings, ahead of TensorFlow
- Employers hiring Python/AI talent include software houses (10Pearls, Confiz, Systems Limited), telecoms (PTCL, Ufone), and banks (Bank Alfalah)
- Freelance work (Upwork, Fiverr) is a major income path, especially chatbot and AI-automation projects, which is exactly what Month 2 builds

Where This Leads

- AI Application Developer / LLM Engineer, building chatbots, RAG systems, and agents
- Freelance AI Automation Consultant, delivering AI integrations directly to clients
- Python Backend Developer at AI-product companies, the deployment half of this course
- This course is built around one identity: someone who can build, ship, and sell a working AI application



Hour 2: Environment Setup

Installing Python

- Download the latest Python 3 release from python.org
- Windows: check "Add Python to PATH" during installation
- macOS/Linux: Python 3 is often preinstalled, verify the version before installing again

Verifying Your Install

Run these in a terminal (Command Prompt, Terminal, or PowerShell).

```
python --version
# or on some systems:
python3 --version
```

VS Code & Your First Script

- VS Code: a free, lightweight code editor, install the official Python extension
- Create a folder for this course, and one .py file per day's work
- Run a script from the terminal with python filename.py
- We'll deep-dive project structure and tooling in Week 4, for now, keep it simple

Your First Script

Save this as hello.py and run it.

```
print("Hello, Python for AI!")
print("Day 1 of 65. Let's get started.")
```

Why Virtual Environments?

- A virtual environment is an isolated, project-specific copy of Python and its packages
- Without one, installing a package for one project can silently break another
- You'll use one from today onward, we'll go deep on managing them in Week 4

SETUP COMMANDS

Creating & Activating a Virtual Environment

Type these commands yourself and confirm the environment activates.

```
# create a virtual environment named "venv"
python -m venv venv

# activate it, Windows
venv\Scripts\activate

# activate it, macOS / Linux
source venv/bin/activate
```



15-Minute Break

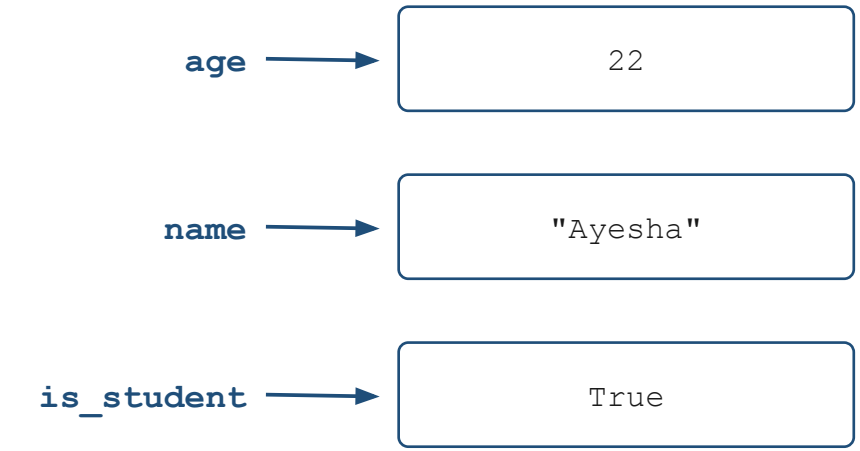
Stretch, grab a coffee. We resume with Hour 3: Python Fundamentals.



Hour 3: Python Fundamentals

What Is a Variable?

- A variable is a named label attached to a value stored in memory
- Python figures out the type automatically, no need to declare it up front
- Variable names are case-sensitive and cannot start with a number
- Use clear, descriptive names: student_marks, not sm or x



name → value stored in memory

Variables & Assignment

The = sign assigns a value to a variable name.

```
name = "Ayesha"
age = 22
is_student = True

print(name, age, is_student)
```

Core Data Types

Every value in Python has a type, check it with type().

```
whole_number = 10           # int
decimal_number = 10.5       # float
text = "Hello"               # str
flag = True                   # bool

print(type(whole_number), type(decimal_number), type(text), type(flag))
```

Arithmetic & Comparison Operators

Arithmetic: + - * / // % ** Comparison: == != > < >= <=

```
a = 10
b = 3

print(a + b, a - b, a * b)
print(a / b)     # true division
print(a // b)    # floor division
print(a % b)     # remainder
print(a > b)     # comparison -> True/False
```

print() and input()

print() shows output, input() reads text typed by the user.

```
name = input("What is your name? ")
print("Hello, " + name + "!")

# f-strings are the cleanest way to mix text and variables
print(f"Hello, {name}! Welcome to the course.")
```

EXAMPLE 1

Practice: Simple Calculator

Type this code yourself and run it.

```
num1 = float(input("Enter first number: "))
num2 = float(input("Enter second number: "))

print("Sum:", num1 + num2)
print("Difference:", num1 - num2)
print("Product:", num1 * num2)
print("Quotient:", num1 / num2)
```

EXAMPLE 2

Practice: Type Conversion

Type this code yourself and run it.

```
age_text = "25"
age_number = int(age_text)

print(type(age_text), type(age_number))
print(age_number + 5)
```

EXAMPLE 3

Practice: Celsius to Fahrenheit

Type this code yourself and run it.

```
celsius = float(input("Enter temperature in Celsius: "))
fahrenheit = (celsius * 9 / 5) + 32
print(f"{celsius}°C is {fahrenheit}°F")
```

EXAMPLE 4, INTERVIEW FAVORITE

Practice: Area & Perimeter of a Rectangle

Type this code yourself and run it.

```
length = float(input("Enter length: "))
width = float(input("Enter width: "))

area = length * width
perimeter = 2 * (length + width)

print(f"Area: {area}")
print(f"Perimeter: {perimeter}")
```

QUESTION 1

Think & Practice

Write a program that asks for two numbers and prints True if the first is greater than the second, without using an if statement (hint: a comparison already gives you True or False).

ANSWER

Answer 1

```
num1 = float(input("Enter first number: "))
num2 = float(input("Enter second number: "))

print(num1 > num2)
```

QUESTION 2, INTERVIEW STYLE

Think & Practice (Harder)

Write a program that asks for an item's price and a discount percentage.

Calculate and print both the amount saved and the final price after the discount.

Use only arithmetic, no if statement needed.

ANSWER

Answer 2

```
price = float(input("Enter item price: "))
discount_percent = float(input("Enter discount percentage: "))

discount_amount = price * discount_percent / 100
final_price = price - discount_amount

print(f"You saved: {discount_amount}")
print(f"Final price: {final_price}")
```

Homework: Practice for Interviews

Try these on your own before next class:

1. Write a program that swaps the values of two variables without using a third variable.
2. Ask for a person's weight (kg) and height (m), calculate and print their BMI.
3. Convert a given number of seconds into hours, minutes, and seconds.
4. Ask for a number and print its square and cube.
5. Ask for two numbers and print the larger one without using if (hint: think about max()).

Key Takeaways

- This course goes deep on one path: Python, then LLM/Agentic AI apps, then shipping and selling them
- A professional setup uses a virtual environment per project, more on this in Week 4
- Every value has a type, Python infers it automatically from what you assign
- input() always returns a string, convert it with int() or float() before doing math

Coming Up: Day 2

- Control Flow
- if / elif / else, boolean logic, nested conditionals
- for and while loops, range(), break/continue/pass
- Common interview patterns: swapping variables, FizzBuzz, and more